FOR AN ALGEBRAICALLY CLOSED RESIDUE FIELD IN CHARACTERISTIC P AND COMPLETE WEAKLY UNRAMIFIED RINGS, WEAK SOLUTIONS EXIST

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof span is the author's complete proof environment at source lines 35385-35511. Its references are recorded as non-load-bearing navigation pointers; both the cross-chapter ones (resolving to the project's tag list) and the same-chapter ones (resolving to their \label lines in the pinned source) concern standard structural facts whose applications stay inside the counted proof span.

Allowed standard prerequisites (editorial).

- Nagata rings and finiteness of integral closures in finite extensions. invoked at proof lines 35387-35388. Is what makes the integral closure of B in a finite extension finite over B, the first step of the proof.
- Valuation rings as maximal with respect to domination, and completeness of a finitely generated module over a complete ring. used at proof lines 35389-35395. Identifies C with the integral closure of B in M and shows the auxiliary rings are complete.
- Inertia and ramification structure theory of extensions of discrete valuation rings. developed earlier in the same section and used throughout the rest of the counted proof span. Supplies the reduction to the totally ramified case in which the characteristic p machinery applies.

Source reference. Original lemma 032W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4035–4035. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00M9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 815–815. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 030M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3953–3953. Cross-chapter navigation pointer into

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Source reference. Original lemma 09DT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11876–11876. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 34037: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34037–34037. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34818: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34818–34818. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 33251: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33251–33251. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34609: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34609–34609. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 33475: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33475–33475. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 35165: sources/stacks-more-algebra/pinned-original-source.tex, source lines 35165–35165. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 35088: sources/stacks-more-algebra/pinned-original-source.tex, source lines 35088–35088. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34676: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34676–34676. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34517: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34517–34517. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 34490: sources/stacks-more-algebra/pinned-original-source.tex, source lines 34490–34490. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original remark 33882: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33882–33882. Same-chapter navigation pointer;

the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \subset B \subset C$ be extensions of discrete valuation rings with fraction fields $K \subset L \subset M$. Assume*

(1) *the residue field k of A is algebraically closed of characteristic $p > 0$,*
(2) *A and B are complete,*
(3) *$A \to B$ is weakly unramified,*
(4) *M is a finite extension of L,*
(5) *$k = \bigcap_{n \geq 1} \kappa_B^{p^n}$*

Then there exists a finite extension K_1/K which is a weak solution for $A \to C$.

Proof. Let M' be any finite extension of L and consider the integral closure C' of B in M'. Then C' is finite over B as B is Nagata by Algebra, Lemma 032W. Moreover, C' is a discrete valuation ring, see discussion in Remark 33882. Moreover C' is complete as a B-module, hence complete as a discrete valuation ring, see Algebra, Section 00M9. It follows in particular that C is the integral closure of B in M (by definition of valuation rings as maximal for the relation of domination).

Let $M \subset M'$ be a finite extension and let $C' \subset M'$ be the integral closure of B as above. By Lemma 34517 it suffices to prove the result for $A \to B \to C'$. Hence we may assume that M/L is normal, see Fields, Lemma 09DT.

If M/L is normal, we can find a chain of finite extensions

$$L = L^0 \subset L^1 \subset L^2 \subset \ldots \subset L^r = M$$

such that each extension L^{j+1}/L^j is either:

(a) purely inseparable of degree p,
(b) totally ramified with respect to B^j and Galois of degree p,
(c) totally ramified with respect to B^j and Galois cyclic of order prime to p,
(d) Galois and unramified with respect to B^j.

Here B^j is the integral closure of B in L^j. Namely, since M/L is normal we can write it as a compositum of a Galois extension and a purely inseparable extension (Fields, Lemma 030M). For the purely inseparable extension the existence of the filtration is clear. In the Galois case, note that G is "the" decomposition group and let $I \subset G$ be the inertia group. Then on the one hand I is solvable by Lemma 33251 and on the other hand the extension M^I/L is unramified with respect to B by Lemma 33475. This proves we have a filtration as stated.

We are going to argue by induction on the integer r. Suppose that we can find a finite extension K_1/K which is a weak solution for $A \to B^1$ where B^1 is the integral closure of B in L^1. Let K_1' be the normal closure of K_1/K (Fields, Lemma 09DT). Since A is complete and the residue field of A is algebraically closed we see that K_1'/K_1 is separable and totally ramified with respect to A_1 (some details omitted). Hence K_1'/K is a weak solution for $A \to B^1$ as well by Lemma 34490. In other words, we may and do assume that K_1 is a normal extension of K. Having done so we consider the sequence

$$L_1^0 = (L^0 \otimes_K K_1)_{red} \subset L_1^1 = (L^1 \otimes_K K_1)_{red} \subset \ldots \subset L_1^r = (L^r \otimes_K K_1)_{red}$$

and the corresponding integral closures B_1^i. Note that $C_1 = B_1^r$ is a product of discrete valuation rings which are transitively permuted by $G = \mathrm{Aut}(K_1/K)$ by Lemma 34609. In particular all the extensions of discrete valuation rings $A_1 \to (C_1)_{\mathfrak{m}}$ are isomorphic and a weak solution for one will be a weak solution for all of them. We can apply the induction hypothesis to the sequence

$$A_1 \to (B_1^1)_{B_1^1 \cap \mathfrak{m}} \to (B_1^2)_{B_1^2 \cap \mathfrak{m}} \to \ldots \to (B_1^r)_{B_1^r \cap \mathfrak{m}} = (C_1)_{\mathfrak{m}}$$

to get a weak solution K_2/K_1 for $A_1 \to (C_1)_{\mathfrak{m}}$. The extension K_2/K will then be a weak solution for $A \to C$ by what we said before. Note that the induction hypothesis applies: the ring map $A_1 \to (B_1^1)_{B_1^1 \cap \mathfrak{m}}$ is weakly unramified by our choice of K_1 and the sequence of fraction field extensions each still have one of the properties (a), (b), (c), or (d) listed above. Moreover, observe that for any finite extension $\kappa_B \subset \kappa$ we still have $k = \bigcap \kappa^{p^n}$.

Thus everything boils down to finding a weak solution for $A \subset C$ when the field extension M/L satisfies one of the properties (a), (b), (c), or (d).

Case (d). This case is trivial as here $B \to C$ is unramified already.

Case (c). Say M/L is cyclic of order n prime to p. Because M/L is totally ramified with respect to B we see that the ramification index of $B \subset C$ is n and hence the ramification index of $A \subset C$ is n as well. Choose a uniformizer $\pi \in A$ and set $K_1 = K[\pi^{1/n}]$. Then K_1/K is a solution for $A \subset C$ by Abhyankar's lemma (Lemma 34037).

Case (b). We divide this case into the mixed characteristic case and the equicharacteristic case. In the equicharacteristic case this is Lemma 34818. In the mixed characteristic case, we first replace K by a finite extension to get to the situation where M/L is a degree p extension of finite level using Lemma 35088. Then the level is a rational number $l \in [0, p)$, see discussion preceding Lemma 35165. If the level is 0, then $B \to C$ is weakly unramified and we're done. If not, then we can replacing the field K by a finite extension to obtain a new situation with level $l' \leq \max(0, l - 1, 2l - p)$ by Lemma 35165. If $l = p - \epsilon$ for $\epsilon < 1$ then we see that $l' \leq p - 2\epsilon$. Hence after a finite number of replacements we obtain a case with level $\leq p - 1$. Then after at most $p - 1$ more such replacements we reach the situation where the level is zero.

Case (a) is Lemma 34676. This is the only case where we possibly need a purely inseparable extension of K, namely, in case (2) of the statement of the lemma we win by adjoining a pth power of the element π. This finishes the proof of the lemma. $\square$